36) A
AB
ABC
ABCD
ABCDE

a_36.c > main()

```c
#include<stdio.h>
int main(){
    for(int i = 1;i <= 5 ; i++)
    {
        for(int j = 1;j<=i;j++)
        {

            printf("%c",j+64);
        }
        printf("\n");
    }
}
```

Problems Output Debug Console Terminal

```
A
AB
ABC
ABCD
ABCDE
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

37)

a_37.c > main()

```c
#include<stdio.h>
int main(){
    for(int i = 1;i <= 5 ; i++)
    {
        for(int k = 5;k > i;k-- )
        {

            printf(" ");
        }
        for(int j = 1;j<=i;j++)
        {

            printf("%d",j);
        }
        printf("\n");
    }
}
```

Problems Output Debug Console Terminal

```
    1
   12
  123
 1234
12345
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

38)

a_38.c > main()

```c
#include<stdio.h>
int main(){
    for(int i = 1;i <= 5 ; i++)
    {

        for(int j = 1;j<=i;j++)
        {

            printf("*");
        }
        printf("\n");
    }
}

```

Problems Output Debug Console Terminal

```
*
**
***
****
*****
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

39)

a_39.c > main()

```c
#include<stdio.h>
int main(){
    for(int i = 1;i <= 5 ; i++)
    {
        for(int j = 1;j<=i;j++)
        {

            printf("%c",i+64);
        }
        printf("\n");
    }
}
```

Problems Output Debug Console Terminal

```
A
BB
CCC
DDDD
EEEEE
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

40)

a_40.c > main()

```c
#include<stdio.h>
int main()
{
    for(int i =1 ; i<= 5 ; i++)
    {
        for( int j= 1; j<=5 ; j++)
        {
            if(i == j) printf("1");
            else printf("0");
        }
        printf("\n");
    }
}
```

Problems Output Debug Console Terminal

```
yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc a_40.c -
10000
01000
00100
00010
00001
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

41)

a_41.c > main()

```c
#include<stdio.h>
int main(){
    int i,j;
    for(int i = 1;i <= 4 ; i++)
    {
        for(int k = 5;k > i;k-- )
        {

            printf(" ");
        }
        for( j = 1;j<=i;j++)
        {

            printf("%d",j);
        }
        for(int j =i-1 ;j>=1 ; j--)
        {
            printf("%d",j);
        }
        printf("\n");
    }
}
```

Problems Output Debug Console Terminal

```
yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" &&
    1
   121
  12321
 1234321
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

42)

a_42.c > main()

```c
#include<stdio.h>
int main(){
    for(int i = 1;i <= 4 ; i++)
    {
        for(int k = 5;k > i;k-- )
        {

            printf(" ");
        }
        for(int j = 1;j<=i;j++)
        {
            printf("*");
        }
        for(int j =i-1 ;j>=1 ; j--)
        {
            printf("*");
        }
        printf("\n");
    }
 for(int s = 1; s<= 3 ; s++)
{
    for(int k = 1 ; k<=s ; k++)
{
    printf(" ");
    if(k==1) printf(" ");
}
for(int u =1 ; u<= 7 - 2*s; u++ )
{
printf("*");
}
printf("\n");
}
}

```

Problems Output Debug Console Terminal

```
yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc a_42.c -o a_42 && "/Users/yagnapatel/Documen
    *
   ***
  *****
 *******
  *****
   ***
    *
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

43) PROGRAM TO RED AND ARRAY OF 10 INTEGER AND PRINT SUM OF NUMBERS.

a_43.c > main()

```c
#include<stdio.h>
int main()
{
    int a[9],sum=0;
    for(int i =0 ;i<9 ;i++)
    {
        printf("Enter the elememnt for array :"); scanf("%d",&a[i]);

            sum += a[i];
    }
    printf("The sum of the element in array is %d",sum);
printf("\n");
}
```

Problems Output Debug Console Terminal

```
Enter the elememnt for array :3
Enter the elememnt for array :1
Enter the elememnt for array :2
Enter the elememnt for array :3
Enter the elememnt for array :4
Enter the elememnt for array :1
Enter the elememnt for array :5
Enter the elememnt for array :1
Enter the elememnt for array :5
The sum of the element in array is 25
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

44) PROGRAM TO READ AN ARRAY OF 10 INTEGER AND COUNT TOTAL NO OF ODD AND TOTAL NO. OF EVERY ELEMENT.

a_44.c > main()

```c
#include<stdio.h>
int main()
{
    int a[9],sum=0,odd=0,even=0;
    for(int i =0 ;i<9 ;i++)
    {
        printf("Enter the elememnt for array :"); scanf("%d",&a[i]);
       if(a[i]%2 == 0) {
        even = even + a[i];
       }
       else {
        odd = odd + a[i];
       }
    }
    printf("Total no. of odd is %d\n",odd);
     printf("Total no. of even is %d\n",even);
}
```

Problems Output Debug Console Terminal

```
Enter the elememnt for array :4
Enter the elememnt for array :2
Enter the elememnt for array :2
Enter the elememnt for array :1
Enter the elememnt for array :2
Enter the elememnt for array :3
Enter the elememnt for array :4
Enter the elememnt for array :5
Enter the elememnt for array :6
Total no. of odd is 9
Total no. of even is 20
yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc a_44.c -o a_44 && "/Us
Enter the elememnt for array :
```

45) PROGRAM TO READ AN ARRAY OF 10 INTEGER AND COUNT TOTAL NO. OF POSITIVE, NEGATIVE, AND ZERO ELEMENTS.

C 45.c > main()

```c
#include<stdio.h>
int main()
{
    int a[9],pos=0,zero=0,neg=0;
    for(int i=0 ;i <=9;i++ )
    {
        printf("Enter the number :");
        scanf("%d",&a[i]);
        if(a[i]>0)
        {
            pos++;
        }
        if(a[i]==0)
        {
            zero++;
        }
        if(a[i]<0)
        {
            neg++;
        }
    }
    printf("The total no. of Positive elements is %d\n",pos);
    printf("The total no. of Negative elements is %d\n",neg);
    printf("The total no. of Zero elements is %d\n",zero);
}
```

Problems Output Debug Console Terminal

```
Enter the number :4
Enter the number :1
Enter the number :2
Enter the number :4
Enter the number :5
Enter the number :
0
Enter the number :-3
Enter the number :7
Enter the number :-2
Enter the number :-9
The total no. of Positive elements is 6
The total no. of Negative elements is 3
The total no. of Zero elements is 1
zsh: abort      "/Users/yagnapatel/Documents/IOP.c/"45
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

46) PROGRAM TO READ N NOS AND FINDOUT MAX AND SECOND MAX.

C 46.c > main()

```c
#include<stdio.h>
int main()
{
    int a[10],max,smax;
    printf("Enter the elememnt for array :"); scanf("%d",&a[0]);
    max = a[0];
    for(int i =1 ;i<10 ;i++)
    {
        printf("Enter the elememnt for array :"); scanf("%d",&a[i]);
    }
     for(int i =0 ;i<10 ;i++){
    if(max < a[i])
       {
        max = a[i];
       }
       for(int i =0 ;i<10 ;i++){
    if(a[i]!=max)
       {
        smax = a[i];
        break;
       }}
        for(int i = 0 ;i<10 ;i++){
     if(smax < a[i] && a[i]!= max )
       {
        smax = a[i];
       }
    }}
    printf("Max is %d\n",max);
    printf("Second max is %d\n",smax);
}
```

Problems Output Debug Console Terminal

```
Enter the elememnt for array :4
Enter the elememnt for array :1
Enter the elememnt for array :2
Enter the elememnt for array :3
Enter the elememnt for array :4
Enter the elememnt for array :1
Enter the elememnt for array :3
Enter the elememnt for array :4
Enter the elememnt for array :
6
Enter the elememnt for array :5
Max is 6
Second max is 5
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

47) PROGRAM TO READ TWO ARRAYS OF 10 INTEGERS AND STORE ADDATION OF THOSE ARRAYS INTO THIRD.

47.c > main()

```c
#include<stdio.h>
int main()
{
   int n;
    printf("Enter n :"); scanf("%d",&n);
     int a[n],b[n],c[n];
    for(int i =1; i<=n ; i++)
    {
        printf("Enter the elements for a:"); scanf("%d",&a[i]);
    }
    for(int i =1; i<=n ; i++)
    {
        printf("Enter the elements for b:"); scanf("%d",&b[i]);
    }
    printf("The addition of both the array is :");
    for(int i =1; i<=n ; i++)
    {
        c[i] = a[i] + b[i];
        printf("%d ",c[i]);
    }
    printf("\n");
}
```

Problems Output Debug Console Terminal

```
Enter n :3
Enter the elements for a:1
Enter the elements for a:2
Enter the elements for a:3
Enter the elements for b:1
Enter the elements for b:2
Enter the elements for b:3
The addition of both the array is :2 4 6
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

48) PROGRAM TO READ TWO ARRAYS OF 10 INTEGERS AND SWAP VALUES THESE ARRAYS.

```c
#include<stdio.h>
int main()
{
   int n;
    printf("Enter n :"); scanf("%d",&n);
     int a[n],b[n],c[n];
    for(int i =1; i<=n ; i++)
    {
        printf("Enter the elements for a:"); scanf("%d",&a[i]);
    }
    for(int i =1; i<=n ; i++)
    {
        printf("Enter the elements for b:"); scanf("%d",&b[i]);
    }
     for(int i =1; i<=n ; i++){
        c[i] = a[i]; a[i]= b[i]; b[i] = c[i];
     }
     printf("Elements of a after swapping :");
    for(int i =1; i<=n ; i++)
    {
        printf(" %d",a[i]);
    }
     printf("\n");
     printf("Elements of b after swapping :");
    for(int i =1; i<=n ; i++)
    {
        printf(" %d",b[i]);
    }
    printf("\n");
}
```

```
Problems  Output  Debug Console  Terminal

Enter n :3
Enter the elements for a:5
Enter the elements for a:36
Enter the elements for a:38
Enter the elements for b:4
Enter the elements for b:85
Enter the elements for b:82
Elements of a after swapping : 4 85 82
Elements of b after swapping : 5 36 38
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

49) PROGRAM TO SORT (SELECTION SORT) AN ARRAY.

```c
int main() {
    int n;
    printf("Enter the size of array :"); scanf("%d",&n);
    int arr[n],i, j, minIndex, temp;

    printf("Enter the elements for array: ");
    for (i = 0; i < n; i++) {
        scanf("%d ", &arr[i]);
    }
    printf("The array is : ");
    for (i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
    for (i = 0; i < n - 1; i++) {
        minIndex = i;
        for (j = i + 1; j < n; j++) {
            if (arr[j] < arr[minIndex]) {
                minIndex = j;
            }

        temp = arr[minIndex];
        arr[minIndex] = arr[i];
        arr[i] = temp;
    }
}

    printf("Sorted array: ");
    for (i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
```

Problems Output Debug Console Terminal

```
Enter the size of array :6
Enter the elements for array: 537
4
72
52
97
64
35
The array is : 537 4 72 52 97 64
Sorted array: 4 52 64 97 72 537
```

50) PROGRAM TO SORT (BUBBLE SORT) AN ARRAY.

```c
#include <stdio.h>
int main() {
int arr[10], n = 10;
printf("Enter 10 integers:");
for (int i = 0; i < n; i++) scanf("%d",&arr[i]);
for (int i = 0; i < n - 1; i++)
 {
    for (int j = 0; j < n - i - 1; j++)
    {
        if (arr[j] > arr[j+1])
        {
            int t = arr[j]; arr[j] = arr[j+1]; arr[j+1] = t;
        }
    }
}
printf("Bubble Sorted:");
for (int i=0; i<n; i++)
printf("%d ", arr[i]); printf("\n");
}
```

```
Problems  Output  Debug Console  Terminal

Enter 10 integers:52
5
2
628
74
47
2
72
73
2
Bubble Sorted:2 2 2 5 47 52 72 73 74 628
yagnapatel@Yagnas-MacBook-Air IOP.c % 6
```

51) PROGRAM TO REVERSE AN ARRAY.

```c
#include<stdio.h>
int main()
{
    int n ;
    printf("Enter the size of array :"); scanf("%d",&n);
    int a[n];
    printf("Enter the elements for array :");
    for(int i = 0 ; i<n ; i++) scanf("%d",&a[i]);
    printf("The array is :");
    for(int i = 0 ; i<n ; i++) printf("%d ",a[i]);
    printf("\n");
    printf("The reverse of array is :");
    for(int i = n-1 ; i>=0 ; i--) printf("%d ",a[i]);
    printf("\n");
}
```

```
Problems  Output  Debug Console  Terminal

Enter the size of array :4
Enter the elements for array :12
34
65
24
The array is :12 34 65 24
The reverse of array is :24 65 34 12
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

52) PROGRAM TO READ A 3*3 MATRIX AND FINDOUT MAX AND MIN ELEMENT.

52.c > main()

```c
#include<stdio.h>
int main()
{
    int a[3][3],i,j,max,min;
    for(i = 0; i<3 ; i++)
    {
        for(j = 0; j<3 ; j++)
        {
            printf("Enter the element for a:"); scanf("%d",&a[i][j]);
        }
    }
    max = min = a[0][0];
    for(i = 0; i<3 ; i++)
    {
        for(j = 0; j<3 ; j++)
        {
            if(a[i][j] > max)
                max = a[i][j];
            if(a[i][j] < min)
                min = a[i][j];
        }
    }
    printf("The max value is %d \n",max);
    printf("The min value is %d \n",min);
}
```

Problems Output Debug Console Terminal

```
Enter the element for a:45
Enter the element for a:362
Enter the element for a:1145
Enter the element for a:23
Enter the element for a:84
Enter the element for a:47
Enter the element for a:23
Enter the element for a:13
Enter the element for a:89
The max value is 1145
The min value is 13
```

53) PROGRAM TO READ A 3*3 MATRIX AND PRINT SUM OF ALL ROWS.

53.c > main()

```c
#include<stdio.h>
int main()
{
int n,i,j;
printf("Enter the count :"); scanf("%d",&n);

int a[n][n],b[n][n],c[n][n],sum=0;
for(i = 0; i < n; i++)
{
    for(j =0 ; j<n;j++){
    printf("Enter the value for a matrix :"); scanf("%d",&a[i][j]);}
}
for( i = 0; i < n; i++)
{
    for(j =0 ; j<n;j++)
    {
        sum += a[i][j];
    }
}
printf("The sum is %d",sum);
printf("\n");
}
```

Problems Output Debug Console Terminal

```
Enter the count :2
Enter the value for a matrix :34
Enter the value for a matrix :63
Enter the value for a matrix :73
Enter the value for a matrix :73
The sum is 243
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

54) PROGRAM TO READ A 3*3 MATRIX AND PRINT ITS TRANSPOSE.

C 54.c > main()

```c
#include<stdio.h>
int main()
{
int n,i,j;
printf("Enter the count :"); scanf("%d",&n);

int a[n][n],t[n][n];
for(i = 0; i < n; i++)
{
    for(j =0 ; j<n;j++){
    printf("Enter the value for a matrix :"); scanf("%d",&a[i][j]);}
}
printf("The matrix after transpose :\n");
     for(i = 0; i < n; i++)
     {
         for(j = 0; j < n; j++)
         {
             t[j][i] = a[i][j];
         }
     }
     for(i = 0; i < n; i++)
     {
         for(j = 0; j < n; j++)
         {
             printf("%d\t",t[i][j]);
         }
         printf("\n");
     }
    }
```

Problems Output Debug Console Terminal

```
Enter the count :2
Enter the value for a matrix :1
Enter the value for a matrix :63
Enter the value for a matrix :73
Enter the value for a matrix :82
The matrix after transpose :
1       73
63      82
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

55) PROGRAM TO READ A 3*3 MATRIX AND ADD THEIR VALUE AND STORE THEM IN THIRD MATRIX.

55.c > main()

```c
#include<stdio.h>
int main()
{
   int n,i,j;
    printf("Enter n :"); scanf("%d",&n);
     int a[n][n],b[n][n],c[n][n];
    for(i =0; i<n ; i++)
    {
        for(j=0;j<n;j++){
        printf("Enter the elements for a:"); scanf("%d",&a[i][j]);
    }
}
    for(i =0; i<n ; i++)
    {
        for(j=0;j<n;j++){
        printf("Enter the elements for b:"); scanf("%d",&b[i][j]);
    }
}
 printf("The addition of both the array is : \n");
    for(i =0; i<n ; i++)
    {
        for(j=0;j<n;j++){
        c[i][j] = a[i][j] + b[i][j];
        printf("%d ",c[i][j]);
    }
    printf("\n");
}
}
```

Problems Output Debug Console Terminal

```
Enter n :2
Enter the elements for a:52
Enter the elements for a:73
Enter the elements for a:73
Enter the elements for a:13
Enter the elements for b:63
Enter the elements for b:90
Enter the elements for b:53
Enter the elements for b:67
The addition of both the array is :
115 163
126 80
vagnapatel@Vagnas-MacBook-Air IOP_c %
```

56) PROGRAM TO READ A 3*3 MATRIX AND SUBTRACT THEIR VALUE AND STORE THEM IN THIRD MATRIX.

C 56.c > main()

```c
#include<stdio.h>
int main()
{
   int n,i,j;
    printf("Enter n :"); scanf("%d",&n);
     int a[n][n],b[n][n],c[n][n];
    for(i =0; i<n ; i++)
    {
        for(j=0;j<n;j++){
        printf("Enter the elements for a:"); scanf("%d",&a[i][j]);
    }
}
    for(i =0; i<n ; i++)
    {
        for(j=0;j<n;j++){
        printf("Enter the elements for b:"); scanf("%d",&b[i][j]);
    }
}
 printf("The subtraction of both the array is : \n");
    for(i =0; i<n ; i++)
    {
        for(j=0;j<n;j++){
        c[i][j] = a[i][j] - b[i][j];
        printf("%d ",c[i][j]);
    }
    printf("\n");
}
}
```

Problems Output Debug Console Terminal

```
Enter n :2
Enter the elements for a:45
Enter the elements for a:78
Enter the elements for a:90
Enter the elements for a:235
Enter the elements for b:13
Enter the elements for b:63
Enter the elements for b:43
Enter the elements for b:12
The subtraction of both the array is :
32 15
47 223
```

57) PROGRAM TO READ A 3*3 MATRIX AND MULTIPLY THEIR VALUE AND STORE THEM IN THIRD MATRIX.

C 57.c > main()

```c
#include<stdio.h>
int main()
{
   int n,i,j;
    printf("Enter n :"); scanf("%d",&n);
     int a[n][n],b[n][n],c[n][n];
    for(i =0; i<n ; i++)
    {
        for(j=0;j<n;j++){
        printf("Enter the elements for a:"); scanf("%d",&a[i][j]);
    }
}
    for(i =0; i<n ; i++)
    {
        for(j=0;j<n;j++){
        printf("Enter the elements for b:"); scanf("%d",&b[i][j]);
    }
}
 printf("The multiplication of both the array is : \n");
    for(i =0; i<n ; i++)
    {
        for(j=0;j<n;j++){
        c[i][j] = a[i][j] * b[i][j];
        printf("%d ",c[i][j]);
    }
    printf("\n");
}
}
```

Problems Output Debug Console Terminal

```
Enter n :2
Enter the elements for a:15
Enter the elements for a:4
Enter the elements for a:678
Enter the elements for a:2
Enter the elements for b:41
Enter the elements for b:35
Enter the elements for b:26
Enter the elements for b:67
The multiplication of both the array is :
615 140
17628 134
```

58) PROGRAM TO PERFORM LINEAR SEARCH ON AN ARRAY.

C 58.c > main()

```c
#include<stdio.h>
int main()
{
    int n,i;
    printf("Enter the array size :"); scanf("%d",&n);
    int a[n],target,flag=0;
    printf("Enter the elements : ");
    for(i = 0 ; i<n ; i++) scanf("%d",&a[i]);
    printf("Enter the element for target :"); scanf("%d",&target);
    for(i = 0 ;i < n ; i++)
    {
        if(a[i] == target ) {flag = 1; break ;}
    }
    if(flag ==1) printf("The target %d is found at %d",target,i+1);
    else printf("The target %d is not found ",target);
    printf("\n");
    }

```

Problems Output Debug Console Terminal

```
Enter the array size :5
Enter the elements : 24
562
56
25
73
Enter the element for target :25
The target 25 is found at 4
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

59) PROGRAM TO READ AN ARRAY AND PRINT THE OCCURRENCE OF ANY PARTICULAR ELEMENT IN THE ARRAY.

C 59.c > ...

```c
#include<stdio.h>
int main()
{
    int n,i;
    printf("Enter the array size :"); scanf("%d",&n);
    int a[n],target,count=0;
    printf("Enter the elements : ");
    for(i = 0 ; i<n ; i++) scanf("%d",&a[i]);
    printf("Enter the element for target :"); scanf("%d",&target);
    for(i = 0 ;i < n ; i++)
    {
        if(a[i] == target ) count++;
    }
    printf("The amount of times target %d come is %d",target,count);
    printf("\n");
    }

```

Problems Output Debug Console Terminal

```
Enter the array size :4
Enter the elements : 13
53
13
67
Enter the element for target :13
The amount of times target 13 come is 2
yagnapatel@Yagnas-MacBook-Air IOP.c %
```